

DAPHNE Electronics Integration

Status update: cold-box mezzanine evidence
DAQ handoff, and PRR gates

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on behalf of the PDS and DAQ groups
Lead, South Dakota - Wednesday, May 20, 2026



What Are We Presenting?

CONFERENCE THESIS

DAPHNE is now an electronics-to-DAQ integration story: mezzanine evidence is strong, new-DAPHNE noise mitigation is the hardware closure item, and PRR readiness depends on frozen data and operations contracts.

1. Mezzanine evidence

2. New DAPHNE gate

3. Integration path

The structure is evidence first, then integration consequence: what was measured, what remains open, and what must be frozen before production review.

Presentation Structure

ONE THROUGHLINE

DAPHNE readiness is the combination of measured analog performance, a closed new-DAPHNE noise gate, and frozen DAQ/PRR contracts.

0. System context

Where DAPHNE sits between cold electronics, firmware data products, and DAQ software.

2. New-DAPHNE hardware gate

Common-mode noise, filter evidence, and the board-spin action needed for closure.

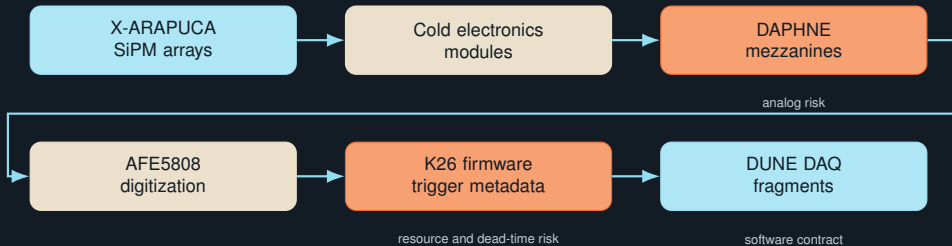
1. Mezzanine evidence package

System under test, module variants, SNR, pulse-shape recovery, and dynamic range.

3. DAQ and PRR closure

Firmware data product, fragment decoding, calibration operations, and remaining ownership.

System Context: Where The Evidence Lands

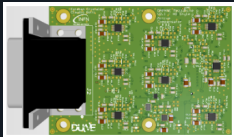
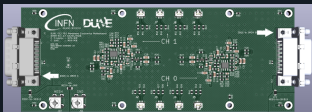


The talk follows this handoff: mezzanines condition the cold-electronics signal, firmware turns it into a data product, and DAQ must make that product usable for monitoring, analysis, and later trigger primitive work.

Mezzanine Evidence: Module Variants

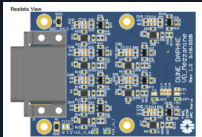
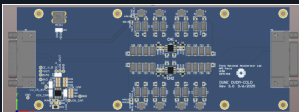
HD

DMEM – FBK membrane



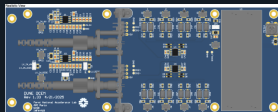
VD

DVDM – HPK membrane



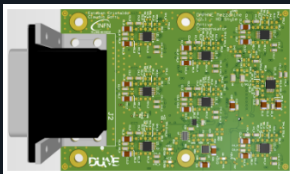
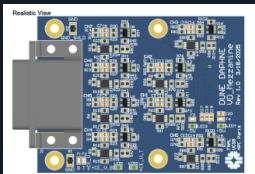
SoF

DCEM – HPK cathode



Integration result: HD, VD, and SoF cold-electronics flavors each have a matched mezzanine integration. The SoF flavor is the one that uses the two-board mechanical split with a flex-cable interface.

Mezzanine Evidence: Functions Under Test



Signal chain

Active differential-to-single-ended conversion followed by AFE compensation for undershoot reduction.

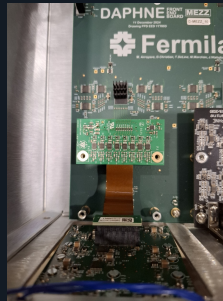
Power protection

Fused +5 V and +3.3 V cold-electronics rails, remote shutdown, power metering, and configurable overcurrent protection.

Flavor coverage

HD, VD, and SoF integrations are represented; the two-board flex-cable mechanical split applies to SoF.

Mezzanine Evidence: Installed Cold-Box Chain



Installed path: front-panel optical input with transimpedance readout and slow-control ADC offset measurements; second board with AFE compensator, flex cable, and DAPHNE interface.

Mezzanine Evidence Map: Three Tests

1A. Signal quality

Cold-box tests in November 2025 and January 2026 showed $\text{SNR} > 4$ for the 1500 P.E. dynamic-range setting.

1B. Pulse recovery

AFE compensation reduced undershoot from around 30% to 0.23%.

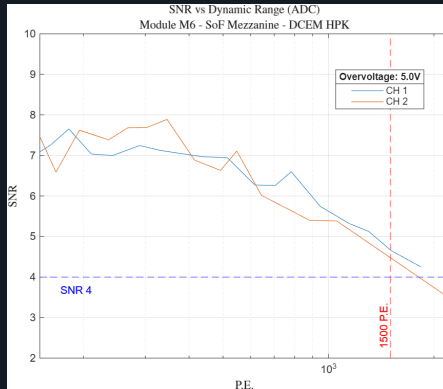
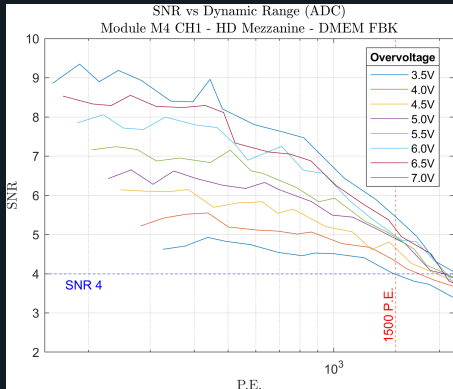
1C. Dynamic range

Membrane and cathode cold-box measurements reached the production target region near 2000 P.E.

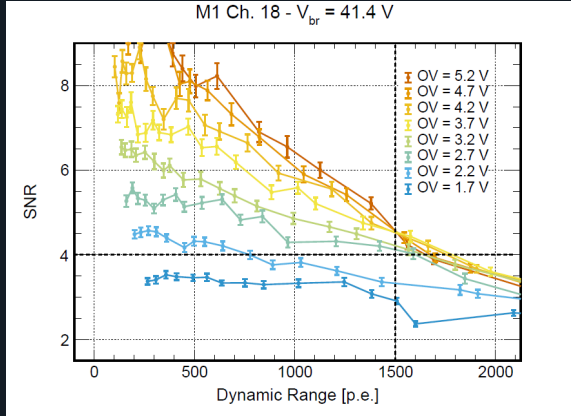
SNR definition used below: SNR versus dynamic range is calculated as $2^{14}/A_{1\text{P.E.}}$. The requirement statement is kept tied to the same module IDs and cold-box conditions as the plots.

Conference use: the next slides are one evidence package, not separate topics: they show that the measured mezzanine chain preserves signal quality, waveform recovery, and usable range.

SNR Result: Cold-Box Curves Clear Threshold



SNR Result: All Modules Clear > 4



Pulse-Shape Evidence: Compensation Method

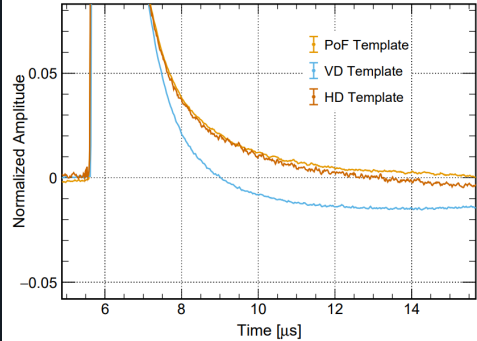
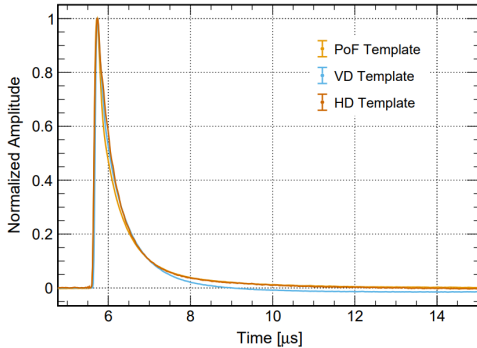
Result: the active differential-to-single-ended stage followed by the AFE compensator reduced undershoot from around 30% to as low as 0.23%.

Why it matters: the pulse-shape correction is not only a cosmetic waveform improvement; it protects baseline recovery and keeps the downstream AFE/DAHPNE path compatible with larger light signals.

Electronics	Rise Time [ns]	Fall Time [ns]	Undershoot [%]
HPK Membrane	62	1240	1.50
FBK Membrane	74	1230	0.47
HPK Cathode	70	1260	0.23

PRR implication: keep the compensation setting, module ID, and residual undershoot table tied to the cold-box evidence packet.

Pulse-Shape Result: Undershoot Reduced To 0.23%



Dynamic-Range Evidence: Measurement Method

Membrane run

During the March 2026 Cold Box run, saturation was measured by increasing LED intensity with DAPHNE set to high front-end attenuation.

Membrane result

FBK at +7 V and HPK at +5 V reached dynamic ranges near 2000 P.E. even at high overvoltage settings.

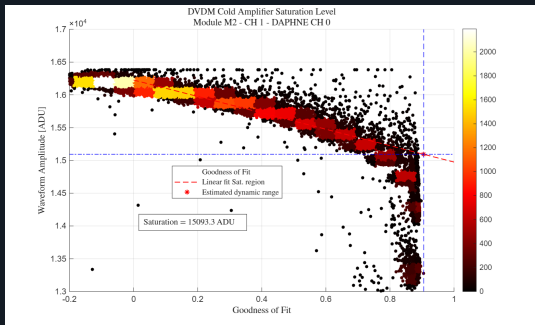
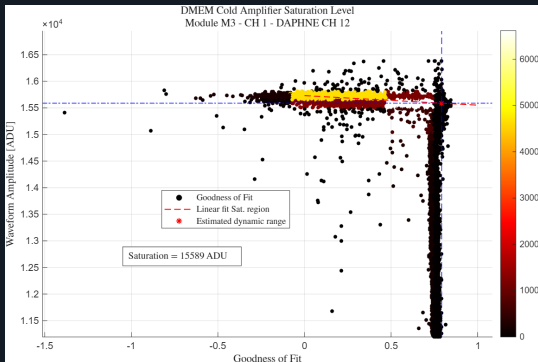
Saturation point

The GoF-vs-amplitude curve defines saturation from the intersection between the saturated plateau and the linear degradation region.

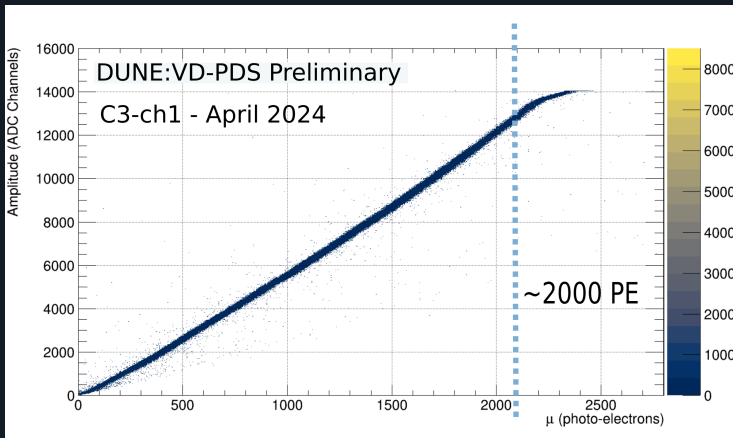
Cathode reference

The April 2024 Cold Box cathode measurement reached 2000 P.E. before saturation.

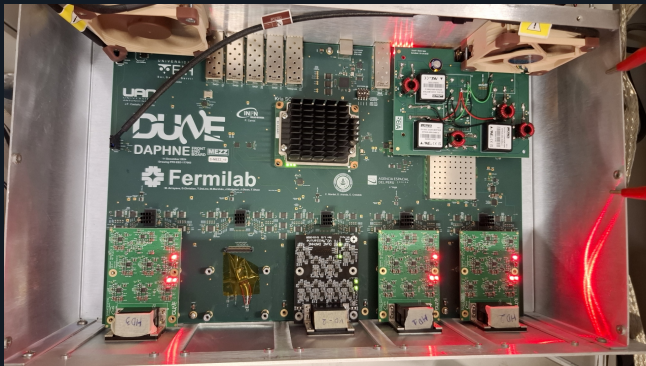
Dynamic-Range Result: Membrane Near 2000 P.E.



Dynamic-Range Result: Cathode Reaches 2000 P.E.



New-DAPHNE Gate: Common-Mode Noise

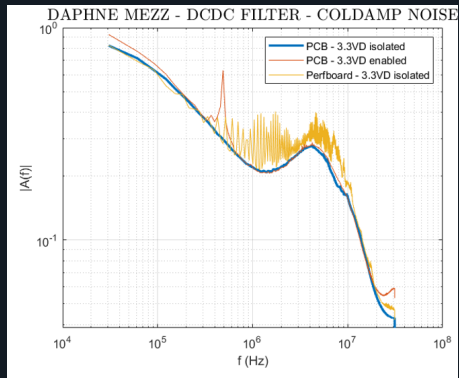
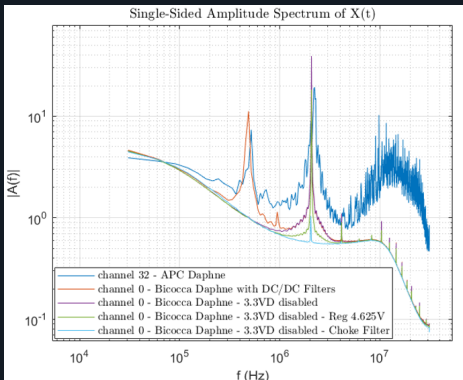


Observed issue: significant common-mode noise, especially on the SoF and HD mezzanines.

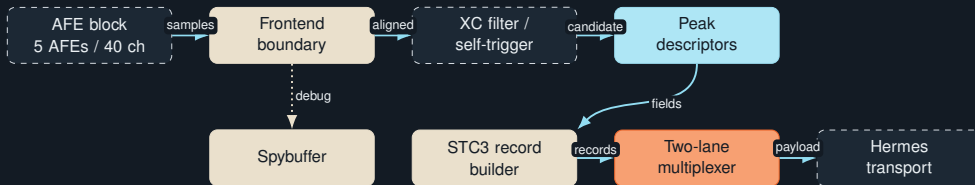
Mitigation path: plug-in filters reduce DC/DC noise and suppress the 2 MHz spike from the -5 V charge pump.

Next board spin: integrate a reduced filter plus an improved -5 V rail for lower noise.

New-DAPHNE Gate: Filter Suppression

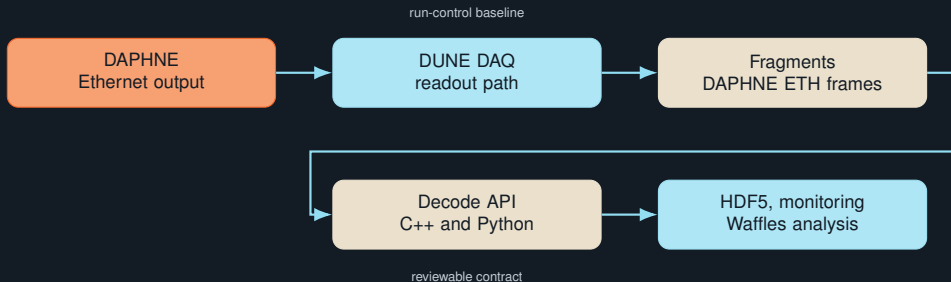


DAQ/PRR Gate: Firmware Data Product



DAQ contract: firmware must make timestamp, channel identity, amplitude, timing, and descriptor validity unambiguous at the fragment boundary. Peak descriptors, x-correlation trigger logic, and dead-time mitigation remain coupled to K26 PL resource margin.

DAQ Integration Baseline



Status: DAPHNE Ethernet data taking is established in DAQ. The remaining production work is to freeze the fragment schema, keep decoding in DUNE-DAQ libraries, and preserve a clean path into HDF5, monitoring, and Waffles analysis.

Calibration Moves Into DAQ Operations



Operational goal: move selected calibration variables from standalone board scripts into `pds-run`, so each scan point has DAQ provenance, run control, monitoring, and reproducible configuration.

PRR Readiness: What Must Be Closed

Area	Conference claim	PRR closure item
Mezzanine performance	■ SNR, pulse shape, and dynamic-range evidence in hand	Attach test matrix, module IDs, and limitations.
Noise mitigation	■ reduced filter path identified	Validate on the next integrated board spin.
Firmware data product	■ descriptor architecture is clear; resource coupling remains active	Freeze fields, rates, and fallback modes.
DAQ integration	■ Ethernet data taking path established	Freeze decode and schema contracts.
Operations and trigger	■ scan recipes and trigger primitive ownership need closure	Assign owners, schedule, and out-of-scope boundary.

Takeaway

WHAT WE CAN SAY

- ▶ Mezzanine analog evidence is strong and presentable.
- ▶ New-DAPHNE noise mitigation has a concrete closure path.
- ▶ DAQ integration is now part of the production story.

WHAT STILL GATES PRODUCTION

- ▶ Integrated-board noise validation.
- ▶ Firmware/DAQ descriptor and decode freeze.
- ▶ Repeatable `pds-run` operations recipes.
- ▶ PRR ownership for remaining trigger primitive scope.

DAPHNE is ready to be presented as an integrated electronics-to-DAQ system, with a short and explicit production closure list.